

Standing Waves on Strings & in Pipes: Harmonics, Wavelengths & Frequencies

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***Standard:** AP Physics 1 (2024 CED) – Unit 8: Mechanical Waves & Sound. SP 6.E: Boundary conditions of strings (fixed-fixed) and pipes (open-open, open-closed) determine allowed harmonics; relate wavelength, frequency, length, and wave speed via $v = f\lambda$.*

EQUATION REFERENCE SHEET

Wave Speed	$v = f\lambda$	v = wave speed, f = frequency, λ = wavelength.
String (both ends fixed)	$\lambda_n = \frac{2L}{n}, \quad f_n = \frac{nv}{2L}$	$n = 1, 2, 3, \dots$ – all harmonics allowed. Nodes at both ends.
Open pipe (both ends open)	$\lambda_n = \frac{2L}{n}, \quad f_n = \frac{nv}{2L}$	$n = 1, 2, 3, \dots$ – same as string. Antinodes at both ends.
Closed pipe (one end closed)	$\lambda_n = \frac{4L}{n}, \quad f_n = \frac{nv}{4L}$	$n = 1, 3, 5, \dots$ – odd harmonics only . Node at closed end, antinode at open end.
Fundamental	f_1 (smallest n)	Higher harmonics: $f_n = n f_1$ (string / open pipe), $f_n = n f_1$ for odd n only (closed pipe).

Worked Example

Read carefully — the same logic applies to every problem below. Identify **boundary type** first, then pick the right formula.

Example — String fixed at both ends

A guitar string of length $L = 0.65$ m is fixed at both ends. The wave speed on the string is $v = 260$ m/s.

- (a) Find the fundamental frequency f_1 .
- (b) Find the frequency of the 3rd harmonic f_3 .
- (c) Find the wavelength λ_3 of the 3rd harmonic.

Solution.

Identify: String fixed at both ends $\Rightarrow$ use $f_n = \frac{nv}{2L}$ with $n = 1, 2, 3, \dots$

$$\text{(a)} \quad f_1 = \frac{(1)(260)}{2(0.65)} = \frac{260}{1.30} = 200 \text{ Hz.}$$

$$\text{(b)} \quad f_3 = 3 f_1 = 3(200) = 600 \text{ Hz.}$$

$$\text{(c)} \quad \lambda_3 = \frac{2L}{n} = \frac{2(0.65)}{3} = \frac{1.30}{3} \approx 0.433 \text{ m.}$$

Check: $v = f_3 \lambda_3 = (600)(0.433) \approx 260$ m/s. $\checkmark$

Practice Problems

Directions: Show all work in the space provided. State the boundary condition, write the relevant equation, then plug in. Round answers to 3 significant figures unless asked otherwise. Take the speed of sound in air to be $v = 343$ m/s unless told otherwise.

1. A string of length $L = 1.20$ m is fixed at both ends. The wave speed on the string is $v = 480$ m/s. Find the fundamental frequency f_1 and the wavelength λ_1 .

2. The fundamental frequency of a string fixed at both ends is $f_1 = 220$ Hz. List the next three harmonic frequencies f_2 , f_3 , f_4 .

3. An organ pipe is **open at both ends** and has length $L = 0.85$ m. Take $v = 343$ m/s. Find the fundamental frequency f_1 and the frequency of the 2nd harmonic f_2 .

4. An organ pipe is **closed at one end** and open at the other, with length $L = 0.85$ m. Take $v = 343$ m/s. Find the fundamental frequency f_1 and the next allowed harmonic frequency.

5. Two pipes have the same length L . Pipe A is open at both ends; Pipe B is closed at one end. Show algebraically that $f_{1,A} = 2 f_{1,B}$, and explain in one sentence why Pipe B has only odd harmonics.

6. A closed pipe (one end closed) produces a fundamental frequency of $f_1 = 196$ Hz when $v = 343$ m/s. Find the length L of the pipe and the wavelength λ_1 .

7. A string fixed at both ends has length $L = 0.50$ m. Its 4th harmonic has frequency $f_4 = 880$ Hz. Find the wave speed v on the string and the wavelength λ_4 .

8. A closed pipe has length $L = 1.10$ m and $v = 343$ m/s. List the first **three allowed** harmonic frequencies (from lowest to highest).

9. A guitar string of length $L = 0.64$ m is tuned so that its fundamental is $f_1 = 330$ Hz (high E). Find the wave speed v on the string and the wavelength λ_2 of the 2nd harmonic.

10. A pipe of length L is observed to resonate at frequencies $f = 250$ Hz and $f = 750$ Hz, but **not** at 500 Hz. Take $v = 343$ m/s.

(a) Is the pipe open at both ends, or closed at one end? Justify.

(b) Find the length L .